Worksheet — Hit the Target Simulation

Use with *Simulation - Hit the target.html* · A.1 Kinematics: plan a projectile to pass through a given point; two possible angles; goes with the cup challenge (place the cup where the ball will land)

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Name: _____ Class: _____ Date: _____

How to use this worksheet

Scan the QR code or open **school.taneski.com/physics/dp/14-hit-the-target** in a browser (offline: *Simulation - Hit the target.html*). Choose a target, set its distance and height, then the launch speed u, angle θ and height h. The **shot planner** shows the working for your settings. ▶ **Shoot**.

Part 1 before the lab (≈ 15 min) · **Part 2** the cup challenge (≈ 15 min) · **Part 3** homework (≈ 30 min).

Plan: horizontal $t = D / (u \cos\theta)$; vertical $y = h + u \sin\theta \cdot t - \frac{1}{2}gt^2$. The ball hits when y equals the target height. $g = 9.81$ m s^{-2}.

Scan to open

Part 1 — Pre-lab: predict, then test

1. A cup is on the floor 8.00 m away. Calculate the launch speed needed at 45° from the ground. Test it.

2. At 10 m s^{-1} (faster than needed at 45°), how many angles land in the cup? Predict, then find them with the help switch on.

3. You launch horizontally from a 0.90 m bench at a cup on the floor 1.20 m away. Calculate the speed needed.

4. The cup is moved to 2.40 m (same bench). How does the needed speed change? Explain.

Part 2 — During the lab: the cup challenge

Roll the ball down the ramp and off the bench. Measure the launch speed u (light gate, or $u = R/t$ from a trial run). Calculate where to put the cup, place it, and release the ball **once**.

h / m	how u was measured	u / m s^{-1}	$t = \sqrt{(2h/g)}$ / s	$D = ut$ / m	result (hit / miss by … cm)

1. Simulation: u = 1.50 m s^{-1} from a 0.90 m bench. Where should the cup go?

2. The same ramp on a 1.20 m table. Where should the cup go now?

3. Your ball landed short or long of the cup. Calculate the percentage error in D and give the most likely reason.

Part 3 — Homework: analyse

1. A basketball is released 2.00 m above the floor at 50° above the horizontal, 4.00 m from the hoop (3.05 m high). What speed is needed?

2. How long does the ball take to reach the hoop?

3. Is the ball rising or falling when it reaches the hoop? Show how you know.

4. A ball is thrown from the ground at 15 m s^{-1} towards a window 5.00 m up in a wall 12.0 m away. Find the two launch angles that hit the window centre.

5. Which of the two shots in Q4 arrives sooner? Explain.

6. A football is kicked from the ground at 18 m s^{-1}, 30° above the horizontal, towards a goal 25 m away (crossbar 2.44 m). How high is the ball at the goal line? Is it a goal?